

Generation and Characterization of Discrete Vortical Gust

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Discrete vortical gusts with different strengths and trajectories were generated and characterized using Time Resolved Particle Image Velocimetry (TR-PIV) in the University of Dayton Water Tunnel (UD-Wat). The discrete vortical gusts were generated by rapidly pitching a flat plate about its quarter chord. The gust profile was tailored by controlling the amplitude of the gust generator. TR-PIV was performed 5 chord lengths downstream of the gust generator to characterize the evolution, growth, and strength of the isolated vortex as it traversed through the field of view. The changes in circulation across the vortex were measured and compared against the Lamb-Oseen vortex model.

Nomenclature

A	=	Oscillation amplitude	S	=	Shear
u	=	x component of velocity	Γ	=	Circulation
U_∞	=	Free stream velocity	Abbreviations		
v	=	y component of velocity	FOV	=	Field of View
t^*	=	Convective time	TR-PIV	=	Time Resolved Particle Image
$\dot{\alpha}$	=	Pitch Rate	Velocimetry		
$\dot{\alpha}_N$	=	Non-dimensional Pitch Rate	UD-WaT	=	University of Dayton Water Tunnel
Q	=	Q-Criterion			
ω	=	Vorticity			

I. Introduction

The study of gusts has grown into an important area of interest in aerodynamics. Small-scale, low-altitude, and low-speed vehicles largely fly under unsteady conditions [1]. Aerodynamic disturbances in unsteady flow can be categorized as streamwise, transverse, or vortical gusts.

Several investigations mentioned in the literature hint at closed-loop control being much more effective in gust mitigation of various types (transverse, streamwise, vortical, etc.) than open-loop control but didn't have the means to implement the closed-loop control in a lab environment. In a study performed by Angulo et al. [2], two transverse gust profiles, a sine squared profile and a top hat profile were used to affect a flat plate being towed through a water tank. The Küssner model [3] was used to predict the plate's response for both gust profiles. In the case of the top-hat gust, the Küssner model underpredicted the lift force at higher gust ratios, and in the case of the sine square profile, the model overpredicted the lift force over the range of gust ratios except between 2 to 3. The deviations from the Küssner model have been attributed to the vorticity shed from the wing and the vorticity present in the gust itself which is referred to as gust force. It is clear that the wing's response to gusts greatly differs depending on the type of gust profile. While an open-loop approach to gust mitigation is explored in detail by various researchers, it also hints that a closed-loop control is a simpler alternative for gust mitigation.

Sedky et al. [4] showed the advantages of using closed-loop control in gust mitigation over open-loop control. Similar to Andreu Angulo et al. [2], Sedky et al. [4] performed a transverse gust study using a sinusoidal gust profile.

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When the wing encounters a sinusoidal gust profile, it is subjected to a non-uniform velocity field which can be decomposed as a sum of the virtual angle of attack and a virtual camber. Using this decomposition, a modified Goman–Khrabrov (GK) Lift Model was used to derive an expression for the effective angle of attack. Using the modified GK model, a closed loop control simulation and an experimental open loop control were employed for gust mitigation. The results from the closed-loop control simulation study were quite impressive for the period when the test article was encountering the gust. However, the experimental open loop control failed to mitigate the effects of lift after the gust encounter. This deviation was accredited to the inability of the model to capture the effects of the leading and trailing edge vortices. This study pointed to the need for implementing closed-loop control for effective gust mitigation and to characterize the system response.

Yeo et al. [5] developed a closed-loop controller for a small, multi-mode aircraft that operates in both helicopter and fixed-wing modes to control the vehicle in gusty environments and during aggressive maneuvers. A nonlinear controller using onboard flow measurements to feedback-linearize the system was experimentally demonstrated to mitigate the effect of vertical gusts on pitch and heave motions. Most quadrotor-type UAVs are inherently unstable and often a PID controller is needed to stabilize the altitude and the Euler angles of the system [6]. Murua et al. [7] simulated the gust load alleviation and trajectory regulation of HALE (High Altitude Long Endurance) flexible aircraft using PID control. A simple PID has been shown to perform well in substantially bringing down the bending strains and providing the necessary guidance authority. Asgharnia et al. [8] simulated the use of RBF-based fractional-order PID (FOPID) to control the pitch angle of a 5 MW wind turbine blade to mitigate mechanical loads.

Mongin et al. [9], [10] developed a closed-loop controller based on PID, state-space, and state-space with Kalman filter to effectively track C_L of a flat plate at large amplitudes and at reduced frequencies of up to 0.3. The angle of attack trajectories taken by the controller was non-sinusoidal for a sinusoidal time-varying C_L reference that was fed into the controller. The study indicated that the PID controller was aggressive but not robust. The state-space controller on the other hand was robust but couldn't track the C_L reference accurately as the PID controller.

Killian et al. [11] used the same PID controller developed by Mongin et al. [10] to mitigate the external disturbances on the wing generated by a periodic vortical gust. The wing controlled by a PID controller was 80% effective in mitigating the unwanted hysteresis and large transients caused by vortex impingement on the flat plate. While the controller was globally effective in mitigating the periodic gust, it struggled to mitigate lift peaks caused by sudden changes in effective angles of attack. By studying the gust entry and exit transients, the controller's bandwidth can be increased to improve the mitigation through a closed-loop controller.

As such, this study will focus on isolating and characterizing a discrete vortical gust that can be used to perform gust entry and exit transient studies. Before performing closed-loop control studies, the discrete vortical gust needs to be characterized in the University of Dayton Water Tunnel. Vadher and Babinsky [12] showed that rapidly pitching a symmetric airfoil can generate gusts of different strengths according to the amplitude and pitch rate of the airfoil. A similar approach will be used in this present study where a flat plate will be pitched at a high pitch rate at three different amplitudes to generate a coherent vortical structure that convects downstream where a wing will be placed. Time-Resolved Particle Image Velocimetry (TR-PIV) was used to characterize the evolution, growth, and strength of the gust. The changes in circulation across the vortex core were measured and compared against the Lamb-Oseen vortex model [13]. A follow-up study will be conducted where the discrete vortical gust will be allowed to impinge on the wing to determine the gust entry and exit transients and the closed-loop controller's performance in mitigating the lift peaks caused by the vortex gust.

II. Experimental Setup

A. University of Dayton Water Tunnel

All experiments were conducted in the University of Dayton Water Tunnel shown in Figure 1. The facility has a 6:1 contraction and a 38 cm wide and 46 cm high test section of 152 cm in length. The speed range of this facility is 0.075 m/s to 0.38 m/s. For this study, all experiments were conducted at a freestream velocity of 0.28 m/s at a Reynolds number of 24,119. The gust generator is operated with an Applied Motion Products HT23-598DC-ZAC stepper motor upstream from the PIV window.

B. Vortical Gust Generator

The vortical gust generator is a flat plate with a 3in. chord and a span of 18.5in. The plate has a thickness of 1/8in. The gust generator is supported on its bottom by a shaft held in place by a magnetic post. The generator spans from the magnetic post to the stepper motor above the free surface of the tunnel as shown in Figure 2b. The gust

generator was placed 5 chords upstream from the window of interest. The distance was chosen to minimize the effect of the wake from the gust generator while also maintaining a strong, distinct gust. The gust generator was commanded a rapid change in pitch angle to three separate amplitudes at nondimensional pitch rate of 45° to create gusts of different strengths. The nondimensional pitch rate (α_N) is defined in Equation 1 where $\dot{\alpha}$ is the dimensional pitch rate, c is the chord, and U_∞ is the freestream velocity. The dimensional pitch rate to achieve a non-dimensional pitch rate of 45° based on chord and freestream was $165^\circ/\text{s}$. The schematic of the gust generator at the corresponding variables is annotated in Figure 2a.

$$\alpha_N = \frac{\dot{\alpha} c}{U_\infty} \quad (1)$$



Figure 1 University of Dayton Water Tunnel Facility

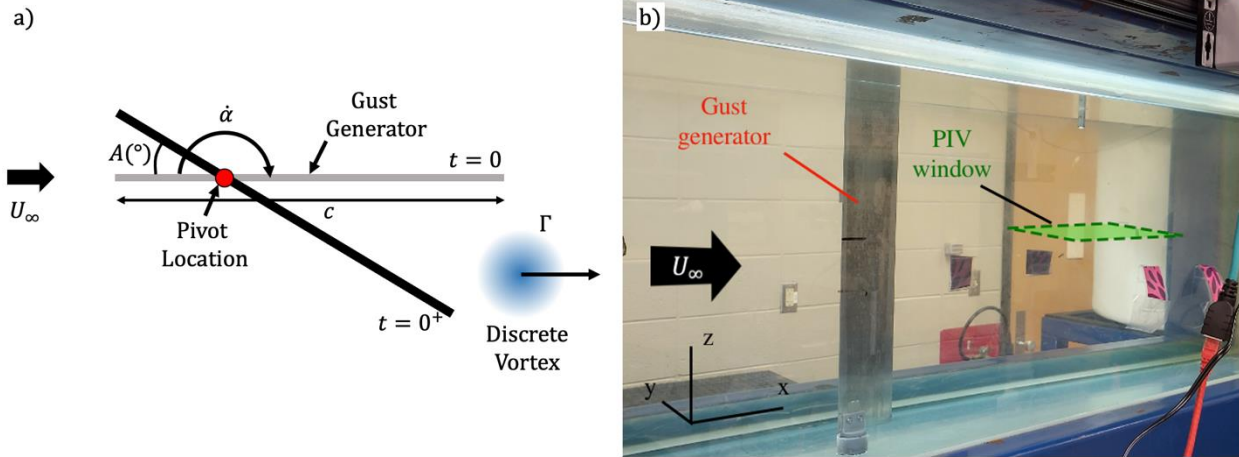


Figure 2 Gust Generator and PIV Window

C. Particle Image Velocimetry

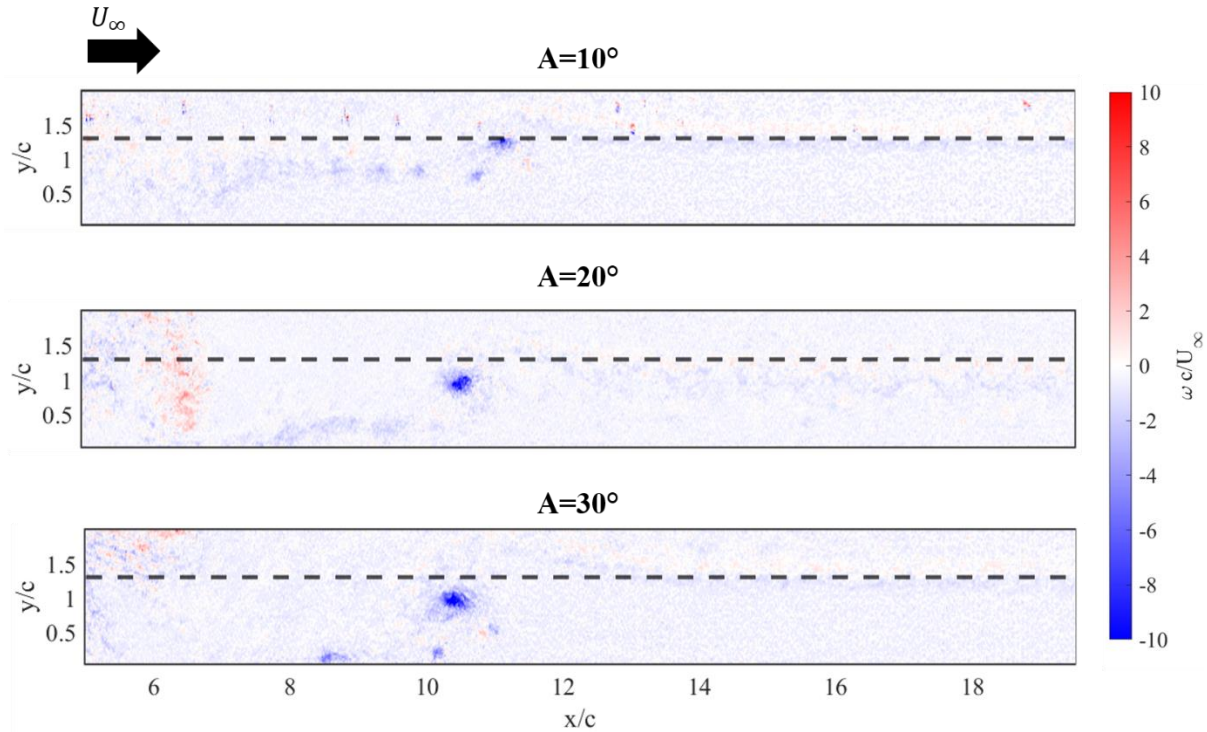
A PCO dimax highspeed camera was used in the TR-PIV study. The field of view is 5 chords downstream of the flat plate gust generator. The tunnel was seeded with $60\mu\text{m}$ polyamide particles and were illuminated with a 532nm LaVision continuous laser. All PIV experiments were conducted for 5 runs and ensemble averaged to reduce noise and uncertainty in the PIV data. The gust generator was pitched to a series of amplitudes at a constant non-dimensional pitch rate according to Table 1.

Table 1 Test Matrix for Discrete Vortical Gust Generation and Characterization

Case	A ($^{\circ}$)	α_N ($^{\circ}$)
1	10	45
2	20	45
3	30	45

III. Results

Coherent vortices of different circulation strengths were generated using the gust generator at various pitch amplitudes A . Figure 3 shows the gust canvases which have been reconstructed by spatially unwrapping the vorticity contours from each PIV frame. Approximately 9 chord lengths of relatively undisturbed flow exists before the vortex on the right side where only the wake from the gust generator is present. On the left side after the vortex there is approximately 3 chord lengths of relatively clean flow before the shedding from the gust generator starts to emerge. The discrete vortex generated with an amplitude of 30° has higher vorticity when compared to the other two cases. The pitch-ramp-hold maneuver used to generate the vortical gust will be tailored to pitch-ramp-pitch-down maneuver to avoid the shedding from the gust generator in the follow-up study.

**Figure 3 Gust canvas of discrete vortices generated at different gust amplitudes**

The two primary objectives of the present study are to determine the vortex trajectory and strength as it travels across the Field of View (FOV). The vortex center was determined through the max Q-criterion **Error! Reference source not found.**, defined in Equation (2) where ω represents the vorticity and S represents shear. The vorticity and shear quantities were calculated using finite differences of the partial derivatives shown in Equations (3) and (4). For most PIV frames, a discrete vortex exists that was easily detectable with Q criterion. The location of maximum Q criterion was recorded as the vortex center for each point in time and treated as the center for future analyses. There were occasional outliers in the x and y locations of the vortex centers. To correct for this, any x or y center outliers were detected and replaced using MATLAB's filloutliers() function. Both x and y were replaced by linearly interpolating between the next nonoutlier values. Outliers in y were detected using a Grubbs test and because x changes more throughout time, outliers in x were detected using a moving median with a window of 10 samples.

$$Q = \frac{1}{2} (\|\omega\|^2 - \|S\|^2) \quad (2)$$

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \quad (3)$$

$$S = \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \quad (4)$$

The vortex trajectories for each of the three gust generator amplitudes are shown in Figure 4. For all cases, the vortex trajectories remain fairly constant as it traverses through the field of view. The trajectory of the vortex in the 20° case convects parallel with the centerline of the field of view. However, the 10° and 30° amplitude cases convect upwards and towards the centerline as the vortex travels across the field of view. As the amplitude of the gust generator deflection increases, the vortex trajectory is offset further from the centerline. This is expected as the higher amplitude will generate a stronger downwash deflecting the vortex further from the centerline.

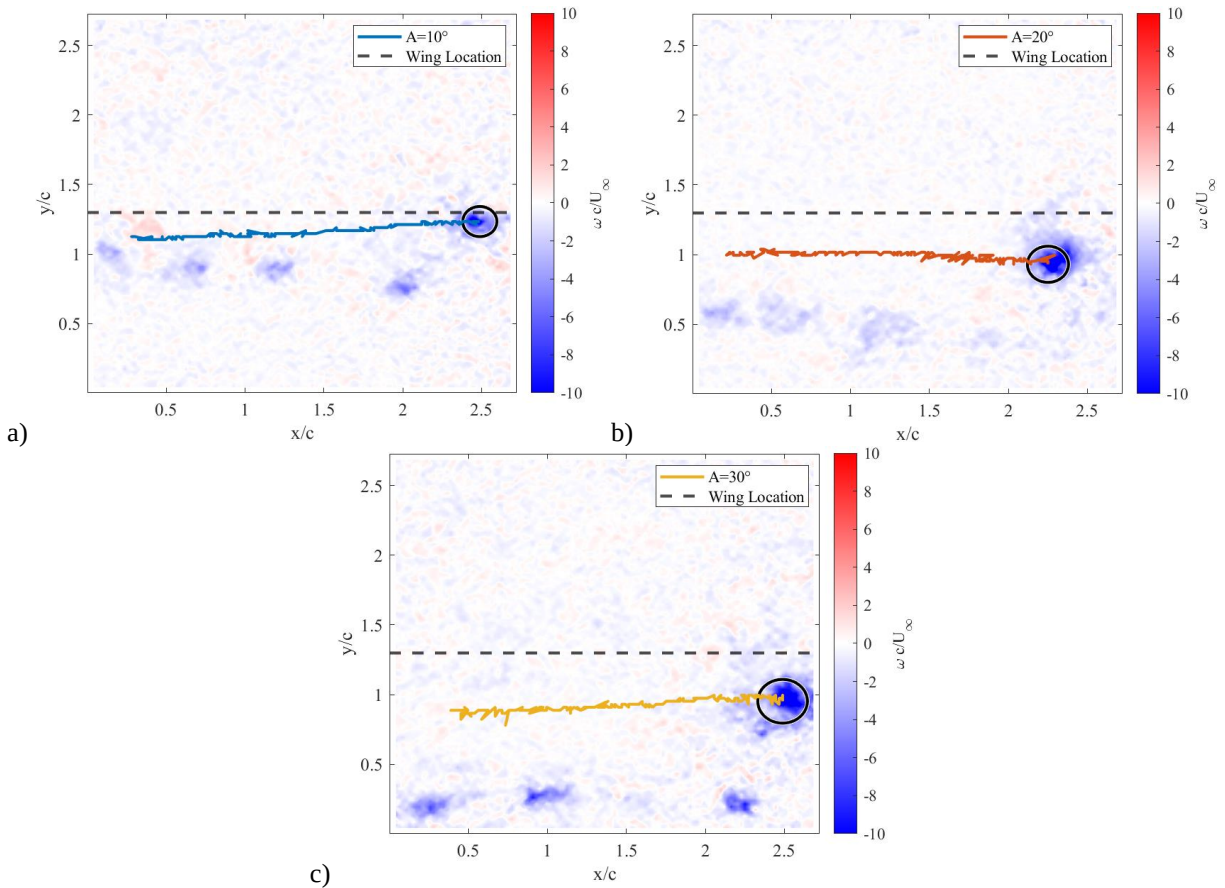


Figure 4 Vortex trajectories in the PIV Field of View for a) $A = 10^\circ$, b) $A = 20^\circ$, c) $A = 30^\circ$,

After the vortex centers were determined, the vortex core radius was found by locating the extrema in the azimuthal velocities of the vortex. Using the previously determined vortex center, the u and v velocities along the vortex x and y axes were obtained. Fig. 3 shows example velocity profiles with the raw velocities taken directly from the ensemble averaged PIV vectors in the dashed lines and the smoothed profiles in solid lines. MATLAB's smoothdata() function was used with 10 sample window on a Savitzky-Golay filter to remove any nonphysical local extrema. The Savitzky-Golay filter's quadratic polynomial fit appeared to best smooth out the curves without attenuating the peaks. The four extrema's (positive and negative u and v) radii were averaged to determine a core radius. For the example shown in Figure 5, an average core radius of 8.25mm was calculated from the radius obtained from u velocity and v velocity as annotated in the figure.

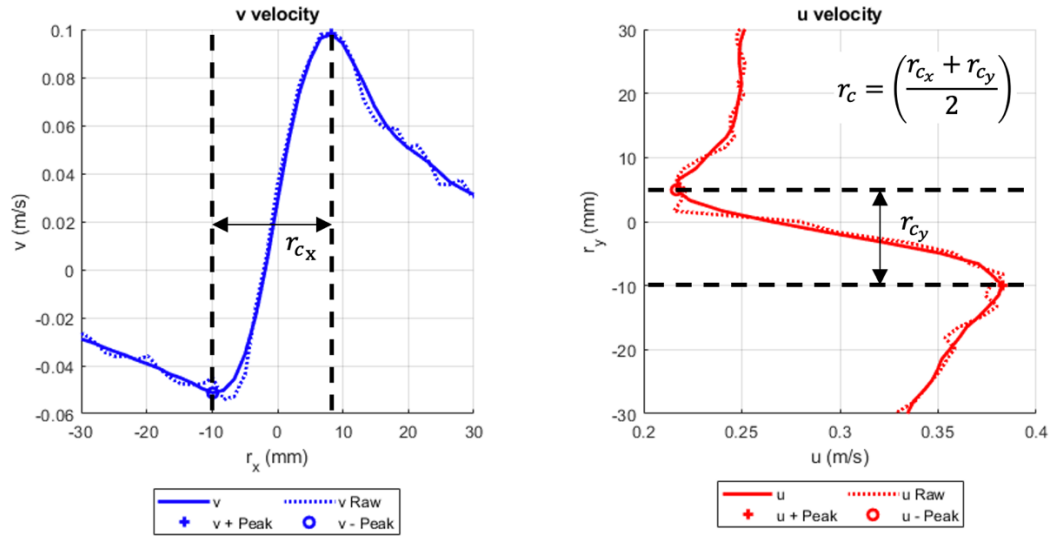


Figure 5 Tangential velocity curves used to calculate the vortex radius

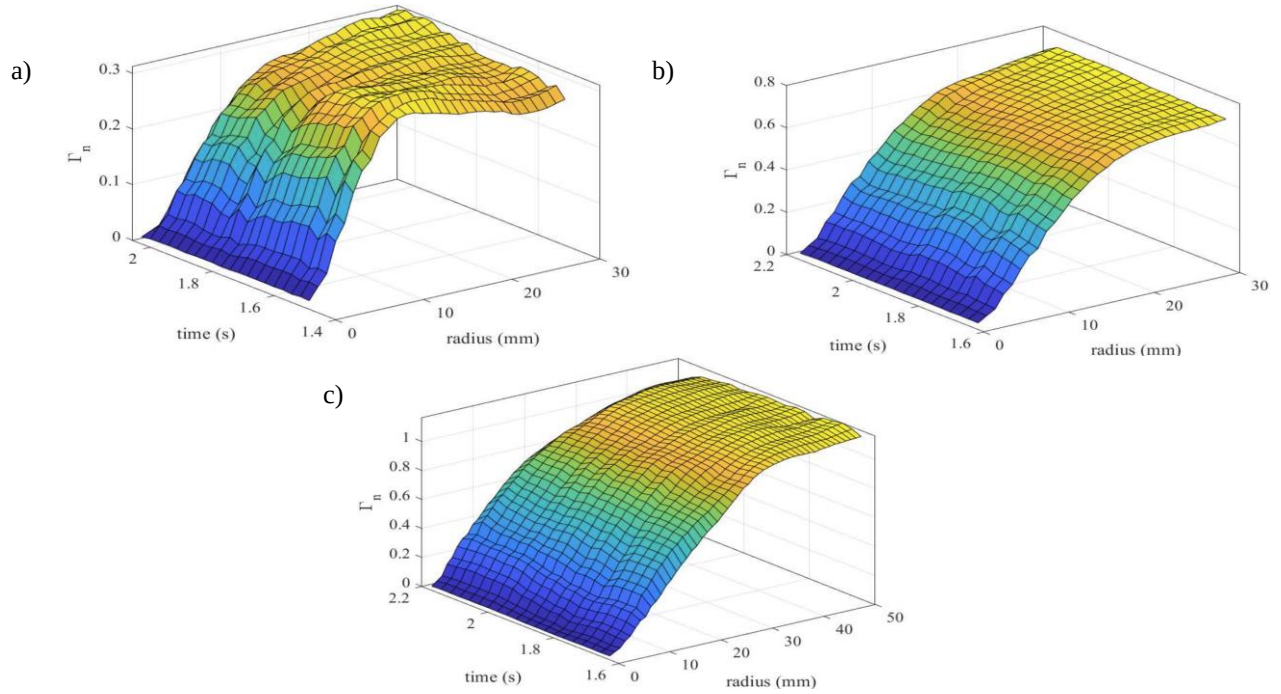


Figure 6 Evolution of vortex circulation with radius and time for a) $A = 10^\circ$, b) $A = 20^\circ$, c) $A = 30^\circ$

The normalized circulation, defined in Equation (5) of the vortex was calculated at each time for an increasing radius from 1-30 mm for the 10° and 20° amplitude cases and 1-50mm for the 30° amplitude case due to its larger diameter. This determined how the circulation grows with radius until it eventually levels off. Figure 6 shows a 3D plot where the surface represents the circulation curve at every point in time for each amplitude. The 20° case is shown in Figure 6b. The 3D plot shows how for this amplitude the circulation tends to reach a constant value of 0.7 at a radius of about 30mm. An ideal Lamb-Oseen circulation profile defined by Equation (6) was fit to the experimental circulation vs radius curve for each point in time by optimizing the Γ_0 term to minimize the RMS error. An example fit is shown in Fig. 7 with an R^2 of 0.99. The optimal Γ_0 term determined for each fit was taken to be the circulation

of the vortex at that point in time. A similar approach was used by Loughnane et al. [14] in determining the circulation of a wingtip vortex.

$$\Gamma_n = \frac{\Gamma}{U_\infty c} \quad (5)$$

$$\Gamma = \Gamma_0 \left(1 - \exp\left(-\frac{r^2}{r_c^2}\right) \right) \quad (6)$$

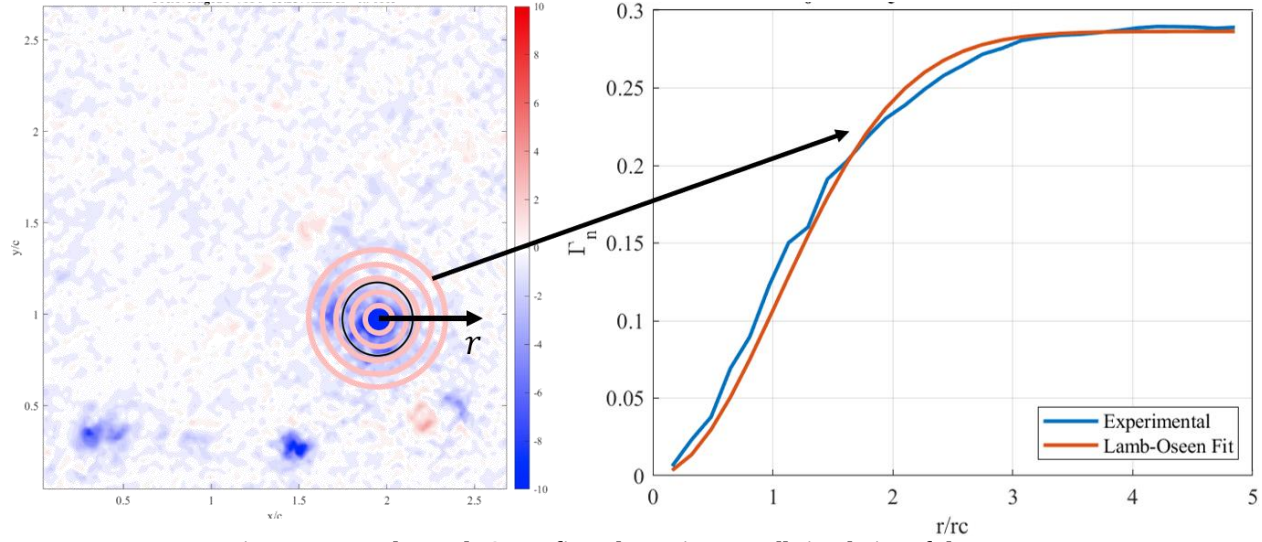


Figure 7 Example Lamb-Oseen fit to determine overall circulation of the vortex

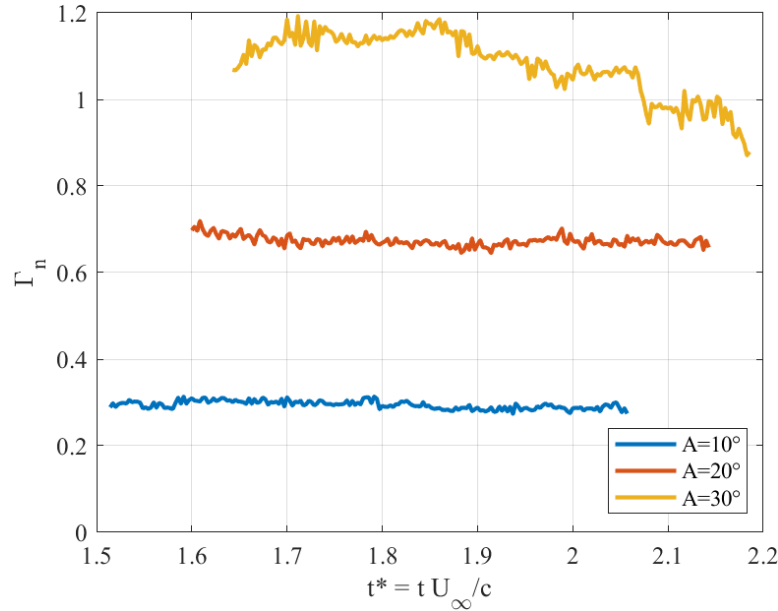


Figure 8 Normalized circulation over time for all amplitude cases

The difference in normalized circulation for each amplitude is shown in Figure 8. The normalized circulation increases with amplitude as expected. The normalized circulation as a function of convective time remains constant for 10° and 20° cases. However, the normalized circulation for the 30° case shows significant variation with convective time when compared to the other two cases. The larger fluctuation in circulation over time stems from secondary vortices around the main vortex shown in Figure 9. The large amplitude of the gust generator produces these secondary structures surrounding the primary vortex core. As the vortices move across the FOV secondary

vortices tend to orbit around the large primary vortex with some even becoming absorbed into it, which could explain the fluctuation seen in circulation shown in Figure 8.

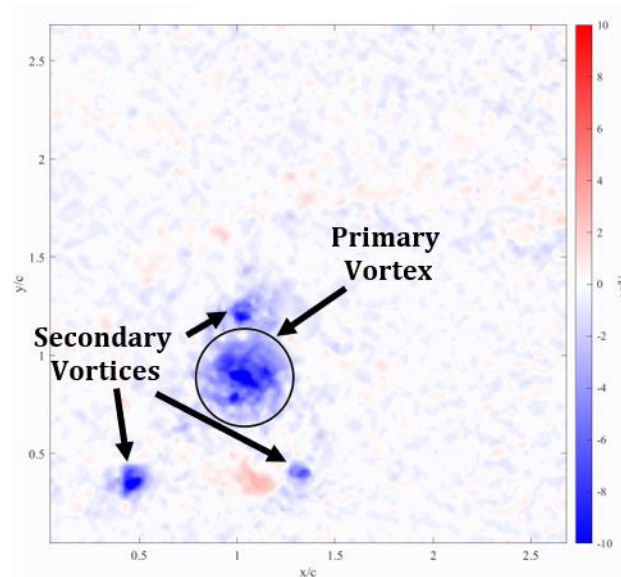


Figure 9 Instantaneous vorticity contour showing the primary and secondary vortices for the $A = 30^\circ$ case

IV. Conclusions

Experimental investigations were conducted with the objective of generating and characterizing a discrete vortical gust. A flat plate gust generator was pitched at a series of amplitudes with a constant non-dimensional pitch rate. The vortex generated from the gust generator was investigated 5 chords downstream. The Q-criterion was used to track the path of the vortex core as it convected downstream, and the circulation of the vortices was determined by fitting a Lamb-Oseen vortex model. The trajectory of the 20° case showed the least drift as it convected downstream, and higher amplitude movements of the gust generator pushed the vortex farther away from the center of the FOV where a wing or flat plate would be located to study a gust encounter. As expected, larger amplitude movements produced higher circulation vortices. In a follow up study, a wing will be placed at close proximity to the vortex and the lift coefficient changes on the plate will be measured. Then the closed loop controller will be used to actively mitigate the changes in the lift coefficient due to the discrete vortex gust.

V. References

- [1] Hufstedler, Esteban AL, and Beverley J. McKeon. "Vortical gusts: experimental generation and interaction with wing." *AIAA Journal* 57, no. 3 (2019): 921-931. <https://doi.org/10.2514/1.J056914>
- [2] Andreu Angulo, Ignacio, Holger Babinsky, Hulya Biler, Girguis Sedky, and Anya R. Jones. "Wing-gust interactions: The effect of transverse velocity profile." In *AIAA Scitech 2020 Forum*, p. 0079. 2020. <https://doi.org/10.2514/6.2020-0079>
- [3] Leishman, Gordon J. *Principles of helicopter aerodynamics with CD extra*. Cambridge university press, 2006
- [4] Sedky, Girguis, Anya R. Jones, and Francis D. Lagor. "Lift regulation during transverse gust encounters using a modified Goman–Khrabrov model." *AIAA Journal* 58, no. 9 (2020): 3788-3798. <https://doi.org/10.2514/1.J059127>
- [5] Yeo, Derrick, Vikram Hrishikeshavan, and Inderjit Chopra. "Gust detection and mitigation on a quad rotor biplane." In *AIAA Atmospheric Flight Mechanics Conference*, p. 1531. 2016. <https://doi.org/10.2514/6.2016-1531>
- [6] Praveen, Viswanadhappalli, and S. Pillai. "Modeling and simulation of quadcopter using PID controller." *International Journal of Control Theory and Applications* 9, no. 15 (2016): 7151-7158.
- [7] Murua, Joseba, Rafael Palacios, and J. M. R. Graham. "Open-loop stability and closed-loop gust alleviation on flexible aircraft including wake modeling." In *53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics*

- and Materials Conference 20th AIAA/ASME/AHS Adaptive Structures Conference 14th AIAA, p. 1484. 2012.
<https://doi.org/10.2514/6.2012-1484>
- [8] Asgharnia, A., A. Jamali, R. Shahnazi, and A. Maheri. "Load mitigation of a class of 5-MW wind turbine with RBF neural network based fractional-order PID controller." *ISA transactions* 96 (2020): 272-286.
<https://doi.org/10.1016/j.isatra.2019.07.006>
- [9] Mongin, Michael P., and Sidaard Gunasekaran. "Lift Regulation Using Closed-Loop Feedback and Control." In *AIAA Scitech 2021 Forum*, p. 2000. 2021. <https://doi.org/10.2514/6.2021-2000>
- [10] Mongin, Michael P., Sidaard Gunasekaran, Albert Medina, Raul Ordonez, and Andrew Killian. "High Amplitude Lift Tracking Using Closed-Loop Feedback and Control." In *AIAA SCITECH 2022 Forum*, p. 0046. 2022. <https://doi.org/10.2514/6.2022-0046>
- [11] Killian, Andrew, Sidaard Gunasekaran, Michael P. Mongin, and Albert Medina. "Periodic Vortical Gust Encounter and Mitigation Using Closed Loop Control." *AIAA SCITECH 2023 Forum*, 2023.
<https://doi.org/10.2514/6.2023-2477>
- [12] Vadher, P., & Babinsky, H. (2023). Experimental Generation and Characterization of Isolated Vortical Gusts. *AIAA SCITECH 2023 Forum* <https://doi.org/10.2514/6.2023-0050>
- [13] Jeong, Jinhee, and Fazle Hussain. "On the identification of a vortex." *Journal of fluid mechanics* 285 (1995): 69-94. <https://doi.org/10.1017/S0022112095000462>
- [14] Loughnane, Faith, Michael Mongin, and Sidaard Gunasekaran. "Effect of airfoil-preserved undulation placement on wing performance and wingtip vortex." *AIAA Journal* 59, no. 5 (2021): 1598-1613.
<https://doi.org/10.2514/1.J059517>